Mathematical Tools for Economists

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1 Vocabulary

Definition 1 (Theorem)**.** A *theorem* is a statement that is true and has been proved to be true.

Definition 2 (Proposition)**.** A statement that is true but not as significant as a theorem is sometimes called a *proposition*.

Definition 3 (Lemma)**.** A *lemma* is a theorem whose main purpose is to help prove another theorem.

Definition 4 (Corollary)**.** A *corollary* is a result that is an immediate consequence of a theorem or proposition.

2 Basic Calculus and Analysis

Definition 5 (Classes of differentiability)**.** Let $U \subseteq \mathbf{R}^n$ be open and let $f : U \to \mathbf{R}^m$. A function f is called continuous if

$$\lim_{\mathbf{y}\to\mathbf{x}} f(\mathbf{y}) = f(\mathbf{x}) \tag{1}$$

for every $\mathbf{x} \in U$. Such a function is said to be of class C^0. A function f is called differentiable if its derivative exists at every point of U. For an integer $k \geq 1$, the function f is said to be of class C^k if all partial derivatives of order at most k exist and are continuous on U. In particular, C^1 describes continuous first derivatives; C^2 describes continuous first and second derivatives, and so on. A function belonging to C^∞ has continuous derivatives of every order and is called smooth.

Definition 6 (Left-hand limit)**.** Let $f : D_f \to \mathbb{R}$, $x \mapsto y = f(x)$, be a function and let $x_0 \in \mathbb{R}$. We say that $f(x)$ converges to $L \in \mathbb{R}$ as x approaches x_0 from the left if, for every sequence $\{x_n\}_{n\in\mathbb{N}}$ with $x_n \in D_f$, $x_n < x_0$, and

$$\lim_{n\to\infty} x_n = x_0, \tag{2}$$

the sequence $\{y_n\}_{n\in\mathbf{N}}$, defined by $y_n = f(x_n)$, converges to L. This is written as

$$\lim_{x\nearrow x_0} f(x) = L \qquad \left(\text{or } \lim_{x\to x_0^-} f(x) = L\right). \tag{3}$$

The number L is called the left-hand limit of f at x_0.

Definition 7 (Right-hand limit)**.** Let $f : D_f \to \mathbb{R}$, $x \mapsto y = f(x)$, be a function and let $x_0 \in \mathbb{R}$. We say that $f(x)$ converges to $R \in \mathbb{R}$ as x approaches x_0 from the right if, for every sequence $\{x_n\}_{n\in\mathbb{N}}$ with $x_n \in D_f$, $x_n > x_0$, and

$$\lim_{n\to\infty} x_n = x_0, \tag{4}$$

the sequence $\{y_n\}_{n\in\mathbf{N}}$, defined by $y_n = f(x_n)$, converges to R. This is written as

$$\lim_{x\searrow x_0} f(x) = R \qquad \left(\text{or } \lim_{x\to x_0^+} f(x) = R\right). \tag{5}$$

The number R is called the right-hand limit of f at x_0.

Definition 8 (Limit)**.** Let $f : D_f \to \mathbb{R}$, $x \mapsto y = f(x)$, be a function and let $x_0 \in \mathbb{R}$. We say that $f(x)$ converges to $y_0 \in \mathbb{R}$ as x approaches x_0 if $f(x)$ converges to y_0 both from the left and from the right. In this case, one writes

$$\lim_{x\to x_0} f(x) = y_0 = \lim_{x\searrow x_0} f(x) = \lim_{x\nearrow x_0} f(x). \tag{6}$$

The number y_0 is called the limit of f as x approaches x_0.

Definition 9 (Differentiable function and its derivative)**.** Let $f : D_f \to \mathbb{R}$, $x \mapsto y = f(x)$, be a function, let $I \subseteq D_f$ be an open interval, and let $x_0 \in I$. The function f is called differentiable at x_0 if the limit

$$\lim_{\Delta x\to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \tag{7}$$

exists. In this case, the limit is denoted by

$$\frac{d}{dx} f(x_0) \tag{8}$$

or

$$f'(x_0), \tag{9}$$

and is called the *difference quotient* or the *derivative* of the function f at x_0.

3 Unconstrained Optimization

Definition 10 (Interior Point)**.** $\mathbf{x}^*$ is an interior point of the domain of F if there exists $r > 0$ such that the whole ball $B_r(\mathbf{x}^*)$ is contained in the domain of F.

Definition 11 (Global and local maxima and minima)**.** Let $F : U \to \mathbf{R^1}$ be a real-valued function of n variables, whose domain U is a subset of $\mathbb{R}^n$.

1. A point $\mathbf{x}^* \in U$ is a **max** of F on U if $F(\mathbf{x}^*) \geq F(\mathbf{x})$ for all $\mathbf{x} \in U$.
2. $\mathbf{x}^* \in U$ is a **strict max** if $\mathbf{x}^*$ is a max and $F(\mathbf{x}^*) > F(\mathbf{x})$ for all $\mathbf{x} \neq \mathbf{x}^*$ in U.
3. $\mathbf{x}^* \in U$ is a **local (or relative) max** of F if there is a ball $B_r(\mathbf{x}^*)$ about $\mathbf{x}^*$ such that $F(\mathbf{x}^*) \geq F(\mathbf{x})$ for all $\mathbf{x} \in B_r(\mathbf{x}^*) \cap U$.
4. $\mathbf{x}^* \in U$ is a **strict local max** of F if there is a ball $B_r(\mathbf{x}^*)$ about $\mathbf{x}^*$ such that $F(\mathbf{x}^*) > F(\mathbf{x})$ for all $\mathbf{x} \neq \mathbf{x}^* \in B_r(\mathbf{x}^*) \cap U$.

If we want to emphasize that a point $\mathbf{x}^*$ is a max of F on the whole domain of U, we call $\mathbf{x}^*$ a **global max** or **absolute max** of F on U.

Theorem 1. *Let $F : U \to \mathbf{R^1}$ be a C^1 function defined on a subset U of $\mathbb{R}^n$. If $\mathbf{x}^*$ is a local max or min of F in U and if $\mathbf{x}^*$ is an interior point of U, then*

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \quad \text{for} \quad i = 1, ..., n. \tag{10}$$

Definition 12 (Critical Point)**.** The n-vector $\mathbf{x}^*$ is a **critical point** of a function $F(x_1, ..., x_n)$ if it satisfies (1).

Theorem 2. *Let $F : U \to \mathbf{R^1}$ be a C^2 function whose domain is a convex open subset U of $\mathbb{R}^n$.*

1. *The following three conditions are equivalent:*
   (a) *F is a concave function of U; and*
   (b) *$F(\boldsymbol{y}) - F(\boldsymbol{x}) \leq DF(\boldsymbol{x})(\boldsymbol{y} - \boldsymbol{x})$ for all $\boldsymbol{x}, \boldsymbol{y} \in U$: and*
   (c) *$D^2F(\boldsymbol{x})$ is negative semidefinite for all $\boldsymbol{x} \in U$*
2. *The following three conditions are equivalent:*
   (a) *F is a convex function on U; and*
   (b) *$F(\boldsymbol{y}) - F(\boldsymbol{x}) \geq DF(\boldsymbol{x})(\boldsymbol{y} - \boldsymbol{x})$ for all $\boldsymbol{x}, \boldsymbol{y} \in U$: and*
   (c) *$D^2F(\boldsymbol{x})$ is positive semidefinite for all $\boldsymbol{x} \in U$*
3. *If F is a concave function on U and $DF(\boldsymbol{x}^*) = \mathbf{0}$ for some $\boldsymbol{x}^* \in U$, then $\boldsymbol{x}^*$ is a global max of F on U.*
4. *If F is a convex function on U and $DF(\boldsymbol{x}^*) = \mathbf{0}$ for some $\boldsymbol{x}^* \in U$, then $\boldsymbol{x}^*$ is a global min of F on U.*

Definition 13 (Convex Set)**.** A set u is a **convex set** if whenever $\mathbf{x}$ and $\mathbf{y}$ are points in U, the line segment joining $\mathbf{x}$ and $\mathbf{y}$,

$$\ell(\mathbf{x},\mathbf{y}) \equiv (t\mathbf{x} + (1-t)\mathbf{y} : 0 \leq t \leq 1, \tag{11}$$

is also in U.

Definition 14. (Definite, semidefinite): Positive definite means x>0; positive semidefinite means $x \geq 0$.

4 Constrained Optimization

Theorem 3. *Let f and h be C^1 functions of two variables. Suppose that $\boldsymbol{x}^* = (x_1^*, x_2^*)$ is a solution of the problem*

$$\max_{x_1, x_2} f(x_1, x_2), \quad \textit{subject to} \quad h(x_1, x_2) = c \tag{12}$$

Suppose further that (x_1^*, x_2^*) is not a critical point of h. Then, there is a real number μ^* such that (x_1^*, x_2^*, μ^*) is a critical point of the Langrangian function

$$L(x_1, x_2, \mu) \equiv f(x_1, x_2) - \mu[h(x_1, x_2) - c]. \tag{13}$$

In other words, at (x_1^*, x_2^*, μ^*)

$$\frac{\partial L}{\partial x_1} = 0, \quad \frac{\partial L}{\partial x_2} = 0, \quad \frac{\partial L}{\partial \mu} = 0. \tag{14}$$

Definition 15 (Binding constraint)**.** If the maximizer $\mathbf{p}$ lies on the boundary of the constraint set $h(x_1, x_2)$, we say the constraint is binding (or is active, effective, or tight) at $\mathbf{p}$.

Definition 16 (Not binding constraint)**.** If the maximizer is in the interior of the constraint set, we say that the constraint is not binding (inactive, ineffective, loose) at $\mathbf{p}$.

5 Homogenous and Homothetic Functions

Definition 17 (Homogenous of degree k)**.** For any scalar k, a real-valued function $f(x_1, ..., x_n)$ is homogenous of degree k if

$$f(tx_1, ..., tx_n) = t^k f(x_1, ..., x_n) \quad \text{for all} \quad x_1, ..., x_n \quad \text{and all} \quad t>0. \tag{15}$$

Theorem 4. *Let $z = f(\boldsymbol{x})$ be a C^1 function on an open cone in $\mathbb{R}^n$. If f is homogenous of degree k, its first order partial derivatives are homogenous of degree $k-1$.*

Definition 18 (Cone)**.** A cone is a set with the property that whenever $\mathbf{x}$ is in the set, every positive scalar multiple $t\mathbf{x}$ of $\mathbf{x}$ is in the set.

Definition 19 (Marginal Rate of Substitution)**.** The marginal rate of substitution (MRS) describes how many units of good Y a consumer is willing to trade for one additional unit of good X. It is the quotient of the marginal utility functions. This means it tells us how much higher the utility of X is compared to that of Y

$$MRS: \quad \left.\frac{dY}{dX}\right|_{\hat{X},\hat{Y}} = -\frac{\partial U(\hat{X},\hat{Y})/\partial X}{\partial U(\hat{X},\hat{Y})/\partial Y} \tag{16}$$

Definition 20 (Marginal Rate of Technical Substitution)**.** The marginal rate of technical substitution (MRTS) is the same as MRS, but with a output-function instead of utility.

Definition 21 (Income Expansion Path)**.** The income expansion path is the parametrized curve $I \mapsto \mathbf{x}(I)$ that indicates the bundle $(x_1, ..., x_n)$ demanded for different income levels.

Theorem 5. *Let $U(\boldsymbol{x})$ be a utility function on $\mathbb{R}^n_+$ that is homogenous of degree k. Then,*

1. *the MRS is constant along the rays from the origin,*
2. *income expansion paths are rays from the origin,*
3. *the corresponding demand depends linearly on income, and*
4. *the income elasticity of demand is identically 1.*

Let $q = f(\boldsymbol{x})$ be a production function on $\mathbb{R}^n_+$ that is homogenous of degree k. Then,

1. *the MRTS is constant along rays from the origin, and*
2. *the corresponding cost function is homogenous of degree $1/k : C(q) = bq^{1/k}$.*

Theorem 6. ***(Euler's Theorem)*** *Let $f(\boldsymbol{x})$ be a C^1 homogenous function of degree k on $\mathbb{R}^n_+$. Then, for all $\boldsymbol{x}$,*

$$x_1 \frac{\partial f}{\partial x_1}(\boldsymbol{x}) + x_2 \frac{\partial f}{\partial x_2}(\boldsymbol{x}) + ... + x_n \frac{\partial f}{\partial x_n}(\boldsymbol{x}) = kf(\boldsymbol{x}), \tag{17}$$

or, in gradient notation,

$$\boldsymbol{x} \cdot \nabla f(\boldsymbol{x}) = kf(\boldsymbol{x}), \tag{18}$$

or, similarly,

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \begin{pmatrix} F_{x1}(\boldsymbol{x}) \\ \vdots \\ F_{xn}(\boldsymbol{x}) \end{pmatrix} = kF(x_1, \ldots, x_n). \tag{19}$$

Theorem 7 (Wicksell's Law)**.** *If f is homogeneous of degree 1 and has diminishing marginal product in x_1, then applying Euler to f_{x1} yields*

$$0 = x_1 f_{x1x1} + x_2 f_{x2x1}$$

$$\Rightarrow f_{x2x1} = -\frac{x_1 f_{x1x1}}{x_2} > 0$$

Hence the marginal product of one factor increases when the other factor increases.

Theorem 8 (Homogenizing a function)**.** *Let $(x_1, ..., x_n) \mapsto f(x_1, ..., x_n)$ be a real-valued function defined on a cone C in $\mathbb{R}^n$. Let k be an integer. Define a new function F of $n+1$ variables by*

$$F(x_1, ..., x_n, z) = z^k \cdot f\left(\frac{x_1}{z}, ..., \frac{x_n}{z}\right). \tag{20}$$

Then, F is a homogenous function of degree k on the cone $C \times \mathbb{R}_+$ in $\mathbb{R}^{n+1}$. Since $f(\boldsymbol{x}) = F(\boldsymbol{x}, 1)$ for all $\boldsymbol{x} \in C$, we can consider f as the restriction of F to an n-dimensional subset of $\mathbb{R}^{n+1}$.

Proposition 3.1 (Radial expansions): The level sets of a homogenous (production) function are radial expansions and contractions of each other.

Definition 22 (Monotonously decreasing function)**.** f is monotonously falling if $f(x) \geq f(x')\forall x, x' \in X$ such that $x \leq x'$.

Definition 23 (Monotonic transformation)**.** Let I be an interval on the real line. Then, $g : I \to \mathbf{R}$ is a **monotonic transformation** of I if g is a strictly increasing function on I. Furthermore, if g is a monotonic transformation and u is a real-valued function of n variables, then we say that

$$g \circ u : \mathbf{x} \mapsto g(u(\mathbf{x})) \tag{21}$$

is a **monotonic transformation of u**. For instance, the functions $3z + 2$, z^2, e^z, and $\ln z$ are all monotonic transformations of $\mathbb{R}_{++}$. Consequently, the utility functions $3xy + 2$, $(xy)^2$, e^{xy}, and $\ln x + \ln y$ are monotonic transformations of the utility function $u(x, y) = xy$.

Definition 24 (Ordinal and cardinal)**.** A characteristic of functions is called ordinal if every monotonic transformation of a function with this characteristic still has that characteristic. Cardinal properties are not preserved by monotonic transformations.

Definition 25 (Homothetic function)**.** A function $v : \mathbb{R}^n_+ \to \mathbb{R}$ is called **homothetic** if it is a monotone transformation of a homogenous function, that is, if there is a monotonic transformation $z \mapsto g(z)$ of $\mathbb{R}_+$ and a homogenous function $u : \mathbb{R}^n_+ \to \mathbb{R}_+$ such that $v(\mathbf{x}) = g(u(\mathbf{x}))$ for all $\mathbf{x}$ in the domain.

Theorem 9. *Let $u : \mathbb{R}^n_+ \to \boldsymbol{R}$ be a strictly monotonic function. Then, u is homothetic if and only if for all $\boldsymbol{x}$ and $\boldsymbol{y}$ in $\mathbb{R}^n_+$,*

$$u(\boldsymbol{x}) \geq u(\boldsymbol{y}) \quad \Leftrightarrow \quad u(\alpha\boldsymbol{x}) \geq u(\alpha\boldsymbol{y}) \quad \forall \quad \alpha > 0. \tag{22}$$

6 Concave and Quasiconcave Functions

Theorem 10 (Convex domain)**.** *Concave and convex functions are required to have convex domains. They contain every line segment between any two points in the set:*

$$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \Rightarrow \theta x_1 + (1 - \theta)x_2 \in C. \tag{23}$$

Definition 26 (Concave and convex n-dimensional functions)**.** A real-valued function f defined on a convex subset U of $\mathbb{R}^n$ is **concave** if for all **x, y** in U and for all t between 0 and 1,

$$f(t\mathbf{x} + (1 - t)\mathbf{y}) \geq tf(\mathbf{x}) + (1 - t)f(\mathbf{y}). \tag{24}$$

A real-valued function g defined on a convex subset U of $\mathbb{R}^n$ is **convex** if for all **x, y** in U and for all t between 0 and 1,

$$g(t\mathbf{x} + (1 - t)\mathbf{y}) \leq tg(\mathbf{x}) + (1 - t)g(\mathbf{y}). \tag{25}$$

Theorem 11. *Let f be a C^1 function on an interval I in $\boldsymbol{R}$. Then, f is concave on I if and only if*

$$f(y) - f(x) \leq f'(x)(y - x) \quad \textit{for all} x, y \in I. \tag{26}$$

The function f is convex on I if and only if

$$f(y) - f(x) \geq f'(x)(y - x) \quad \textit{for all} x, y \in I. \tag{27}$$

Theorem 12. *Let f be a C^1 function on a convex subset U of $\mathbb{R}^n$. Then, f is concave on U if and only if for all $\boldsymbol{x}$, $\boldsymbol{y}$ in U:*

$$f(\boldsymbol{y}) - f(\boldsymbol{x}) \leq Df(\boldsymbol{x})(\boldsymbol{y} - \boldsymbol{x}); \tag{28}$$

that is,

$$f(\boldsymbol{y}) - f(\boldsymbol{x}) \leq \frac{\partial f}{\partial x_1}(\boldsymbol{x})(y_1 - x_1) + ... + \frac{\partial f}{\partial x_n}(\boldsymbol{x})(y_n - x_n). \tag{29}$$

Similarly, f is convex on U if and only if $f(\boldsymbol{y}) - f(\boldsymbol{x}) \geq Df(\boldsymbol{x})(\boldsymbol{y} - \boldsymbol{x})$ for all $\boldsymbol{x}$, $\boldsymbol{y}$ in U.

Definition 27 (Quasiconcavity and Quasiconvexity)**.** Let $f : \mathcal{S} \mapsto \mathbb{R}$.

1. The function f is quasiconcave on $\mathcal{S}$ if and only if the upper level set $C_a^+ = \{x \in \mathcal{S} : f(x) \geq a\}$ is a convex set for every $a \in \mathcal{S}$.
2. The function f is quasiconvex on $\mathcal{S}$ if and only if the lower level set $C_a^- = \{x \in \mathcal{S} : f(x) \leq a\}$ is a convex set for every $a \in \mathcal{S}$.

7 Difference Equations

Definition 28 (Functional Equation)**.** A *functional equation* is an equation where the unknown is the function.

Definition 29 (Ordinary difference equation)**.** An *ordinary difference equation* is a functional equation involving one or more of the differences $\Delta y, \Delta^2 y$, etc. of a function of time.

Definition 30 (Linear, constant-coefficient non-homogenous difference equation)**.** The general n-th order form of such equations is

$$c_n y_{t+n} + c_{n-1} y_{t+n-1} + ... + c_1 y_{t+1} + c_0 y_t = g(t), \tag{30}$$

where the c's are given constants and $g(t)$ is a known function.

Definition 31 (Linear, constant-coefficient homogenous difference equation)**.** The general n-th order form of such equations is

$$c_n y_{t+n} + c_{n-1} y_{t+n-1} + ... + c_1 y_{t+1} + c_0 y_t = 0. \tag{31}$$

Theorem 13. *If $y_1(t)$ is a solution of the homogenous equation, then $Ay_1(t)$, where A is an arbitrary constant, is also a solution.*

Theorem 14. *If $y_1(t), y_2(t)$ are two distinct (linearly independent) solutions of the homogenous equation ($n > 1$), then $A_1 y_1(t) + A_2 y_2(t)$ is also a solution for any two constants A_1, A_2.*

Theorem 15. *If $\overline{y}(t)$ is a particular solution of the non-homogenous equation, the general solution of this same equation is obtained adding $\overline{y}(t)$ to the general solution of the corresponding homogenous equation.*

Theorem 16 (General form of first-order non-homogenous equations)**.** *The general form of these equations is*

$$c_1 y_t + c_0 y_{t-1} = g(t), \tag{32}$$

where c_0, c_1 are given in constants and $g(t)$ is a known function and its general solution is $y_t = A(-b)^t + \overline{y}_t \Leftrightarrow A = \frac{y^* - \overline{y}_{t^*}}{(-b)^{t^*}}$.

Theorem 17 (General form of first-order homogenous equations)**.** *The general form of these equations is*

$$c_1 y_t + c_0 y_{t-1} = 0, \tag{33}$$

or

$$y_t + b y_{t-1} = 0, \tag{34}$$

where $b \equiv c_0/c_1$ and the solution is $y_t = A(-b)^t \Leftrightarrow A = y^*/(-b)^{t*}$.

Theorem 18. *(Monotony and convergence characteristics of the general solution of first-order difference equations with constant coefficients): There are only four possible outcomes: Monotonous and convergent $A \in (0,1)$, oscillating and convergent $A \in (-1,0)$, monotonous and divergent $A \in (1,\infty)$, and oscillating and divergent $A \in (-\infty,-1]$.*

Theorem 19 (First-order systems)**.** *The following is a first order non-homogenous system*

$$\begin{aligned} y_{t+1} &= a_{11}y_t + a_{12}z_t + g_1(t), \\ z_{t+1} &= a_{21}y_t + a_{22}z_t + g_2(t), \end{aligned} \tag{35}$$

where the coefficients a_{ij} are given constants and $g_1(t), g_2(t)$ are known functions. The corresponding homogenous system is

$$\begin{aligned} y_{t+1} &= a_{11}y_t + a_{12}z_t, \\ z_{t+1} &= a_{21}y_t + a_{22}z_t. \end{aligned} \tag{36}$$

We solve (37) by solving the second equation for z_t, creating a duplicate where time is shifted by one period, taking the first equation in (37) and shifting it too and then substituting the z-terms with the ones we just derived. This eventually gives us the second-order difference equation

$$y_t - (a_{11} + a_{22})y_{t-1} - (a_{12}a_{21} - a_{11}a_{22})y_{t-2} = 0 \tag{37}$$

with only one unknown y_t.

Solution 1 (General solution of the homogenous system: Direct method)**.** *We solve the system from equation (36) directly by taking it's homogenous counterpart and then substituting $y_t = \alpha_1\lambda^t, z_t = \alpha_2\lambda^t$. This gives us*

$$\begin{aligned} \alpha_1\lambda^{t+1} &= a_{11}\alpha_1\lambda^t + a_{12}\alpha_2\lambda^t, \\ \alpha_2\lambda^{t+1} &= a_{21}\alpha_1\lambda^t + a_{22}\alpha_2\lambda^t. \end{aligned} \tag{38}$$

This is equivalent to

$$\begin{aligned} \lambda^t[\alpha_1\lambda - a_{11}\alpha_1 - a_{12}\alpha_2] = 0, \\ \lambda^t[\alpha_2\lambda - a_{21}\alpha_1 - a_{22}\alpha_2] = 0, \end{aligned} \tag{39}$$

leading us to

$$\begin{aligned} \alpha_1(\lambda - a_{11}) - a_{12}\alpha_2 = 0, \\ \alpha_2(\lambda - a_{22}) - a_{21}\alpha_1 = 0. \end{aligned} \tag{40}$$

We must invert the sign and take the determinant of the matrix A which represents that consequential system. Remember that the variables α_1, α_2 must be dropped out of the equation.

$$\begin{aligned} det(A) = \begin{vmatrix} (a_{11} - \lambda) & a_{12} \\ a_{21} & (a_{22} - \lambda) \end{vmatrix} \\ \Leftrightarrow (a_{11} - \lambda)(a_{22} - \lambda) - a_{21}a_{12} = 0 \\ \Leftrightarrow a_{11}a_{22} - a_{11}\lambda - \alpha_{22}\lambda + \lambda^2 - a_{21}a_{12} = 0 \\ \Leftrightarrow \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{21}a_{12}) = 0. \end{aligned} \tag{41}$$

The determinantal equation in the first equality and its expanded form are called the characteristic equation of the system of difference equations. From Theorem 18 we know that the constants a_{ij} are known, such that the last equation will give us two solutions λ_1, λ_2.

8 Differential Equations

Theorem 20 (General form of first-order non-homogenous equations)**.** *The general form of these equations is*

$$\begin{aligned} \dot{x}(t) &= a_{11}x(t) + a_{12}y(t) + g(t), \\ \dot{y}(t) &= a_{21}x(t) + a_{22}y(t) + g(t), \end{aligned} \tag{42}$$

where a_{ij} are given constants and $g(t)$ is a known function.

Solution 2 (General solution of the homogenous system: Direct method)**.** *We solve the system from equation (43) directly by taking it's homogenous counterpart and setting* $x(t) = \alpha_1 e^{\lambda t}, y(t) = \alpha_2 e^{\lambda t}$*. This gives us*

$$\begin{aligned} \alpha_1 \lambda e^{\lambda t} &= a_{11}\alpha_1 e^{\lambda t} + a_{12}\alpha_2 e^{\lambda t}, \\ \alpha_2 \lambda e^{\lambda t} &= a_{21}\alpha_1 e^{\lambda t} + a_{22}\alpha_2 e^{\lambda t}. \end{aligned} \tag{43}$$

This is equivalent to

$$\begin{aligned} e^{\lambda t}[(a_{11} - \lambda)\alpha_1 + a_{12}\alpha_2] &= 0, \\ e^{\lambda t}[a_{21}\alpha_1 + (a_{22} - \lambda)\alpha_2] &= 0. \end{aligned} \tag{44}$$

leading us to

$$\begin{aligned} (a_{11} - \lambda)\alpha_1 + a_{12}\alpha_2 &= 0, \\ a_{21}\alpha_1 + (a_{22} - \lambda)\alpha_2 &= 0. \end{aligned} \tag{45}$$

We must take the determinant of the matrix A which represents that consequential system. Remember that the variables α_1, α_2 *must be dropped out of the equation.*

$$\begin{aligned} det(A) = \begin{vmatrix} (a_{11} - \lambda) & a_{12} \\ a_{21} & (a_{22} - \lambda) \end{vmatrix} & \\ \Leftrightarrow (a_{11} - \lambda)(a_{22} - \lambda) - a_{21}a_{12} = 0 & \\ \Leftrightarrow a_{11}a_{22} - a_{11}\lambda - \alpha_{22}\lambda + \lambda^2 - a_{21}a_{12} = 0 & \\ \Leftrightarrow \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{21}a_{12}) = 0. & \end{aligned} \tag{46}$$

The determinantal equation in the first equality and its expanded form are called the characteristic equation of the system of difference equations. From Theorem 18 we know that the constants a_{ij} *are known, such that the last equation will give us two solutions* λ_1, λ_2*.*

9 Optimal Control